

Physico-mechanical properties of soil

- Introduction to soil mechanics and geotechnical engineering
- Origin of soils
- Basic relationships
- **Index properties including consistency and gradation**
- Clay mineralogy
- Classification of engineering soils
- Engineering properties of soils compressibility, consolidation, compaction and strength.

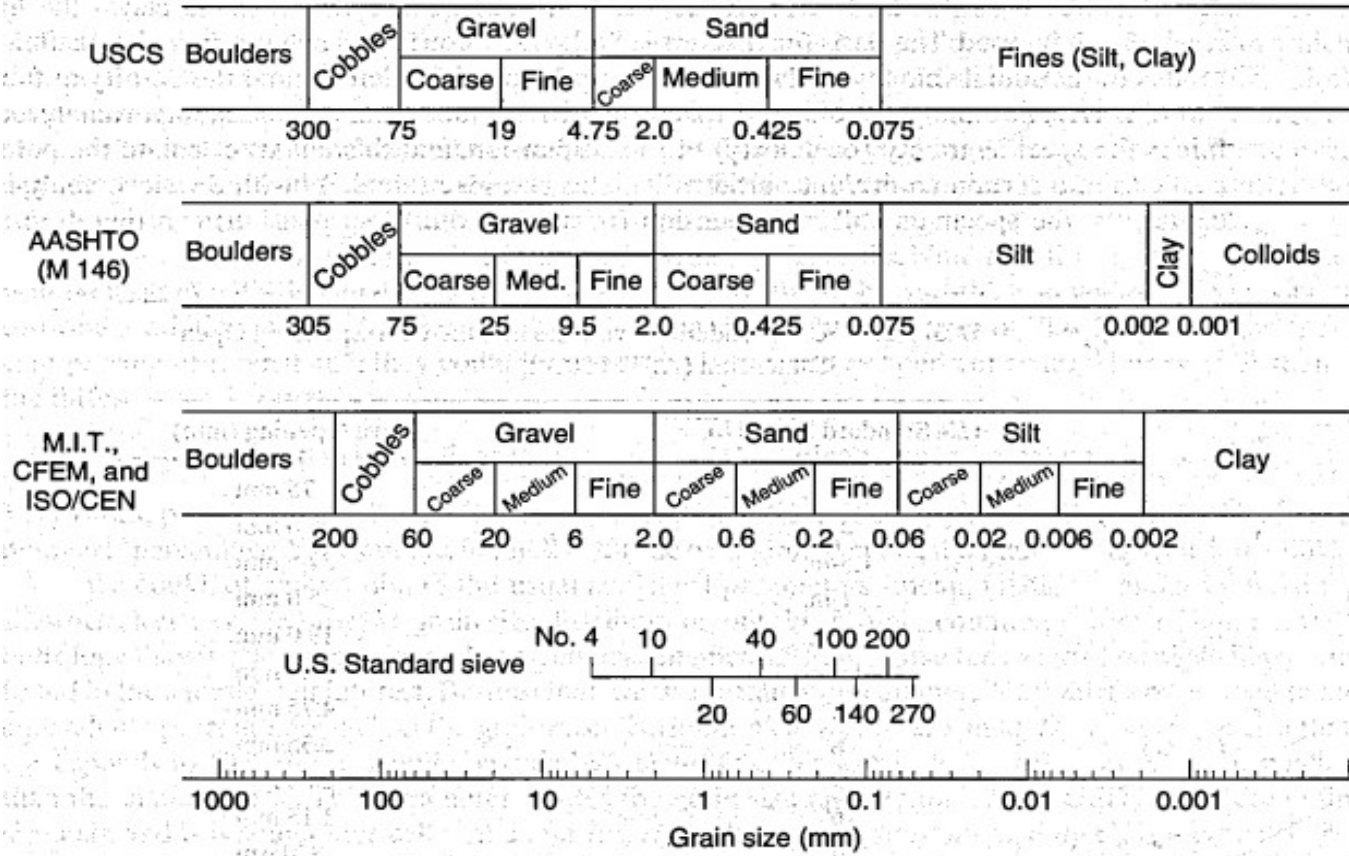
Introduction

- In the field of geotechnical engineering it is necessary to evaluate certain index and engineering properties to predict the engineering behaviour of the soil deposits.
- These properties can be correlated with the strength and compaction characteristics of the soils.

The **texture** of a soil is its appearance or "feel," and it depends on the relative 'sizes and shapes of the particles as well as the range or distribution of those sizes. Thus coarse-grained soils such as sands or gravels obviously appear coarse textured, while a fine-textured soil is composed mainly of very small mineral grains invisible to the naked eye. Silts and clay soils are good examples of fine-textured soils.

	Soil Name		
	Gravels, Sands	Silts	Clays
Grain size	Coarse grained Can see individual grains by eye	Fine grained Cannot see individual grains	Fine grained Cannot see individual grains
Characteristics	Cohesionless Nonplastic Granular	Cohesionless Nonplastic Granular	Cohesive Plastic —
Effect of water on engineering behavior	Relatively unimportant (exception: loose, saturated granular materials and dynamic loadings)	Important	Very important
Effect of grain-size distribution on engineering behavior	Important	Relatively unimportant	Relatively unimportant

GRAIN SIZE AND GRAIN SIZE DISTRIBUTION



USCS = Unified Soil Classification System (U.S. Bureau of Reclamation, 1974; U.S. Army Engineer WES, 1960); ASTM D 2487

AASHTO = American Association for State Highway and Transportation Officials (1998)

M.I.T., C.F.E.M., ISO/CEN = Massachusetts Institute of Technology (Taylor, 1948); Canadian Foundation Engineering Manual (2006); International Standardisation Organisation and Comité Européen de Normalisation.

U.S. Standard Sieve No.**Sieve Opening (mm)**3-in.^a

75 mm

2-in.

50 mm

1.5-in.^a

37.5 mm

1-in.

25.0 mm

0.75-in.^a

19.0 mm

0.375-in.^a

9.5 mm

4^a

4.75 mm

8^a

2.35 mm

10

2.00 mm

16^a

1.18 mm

20

850 μm

30^a

600 μm

40

425 μm

50^a

300 μm

60

250 μm

100^a

150 μm

140

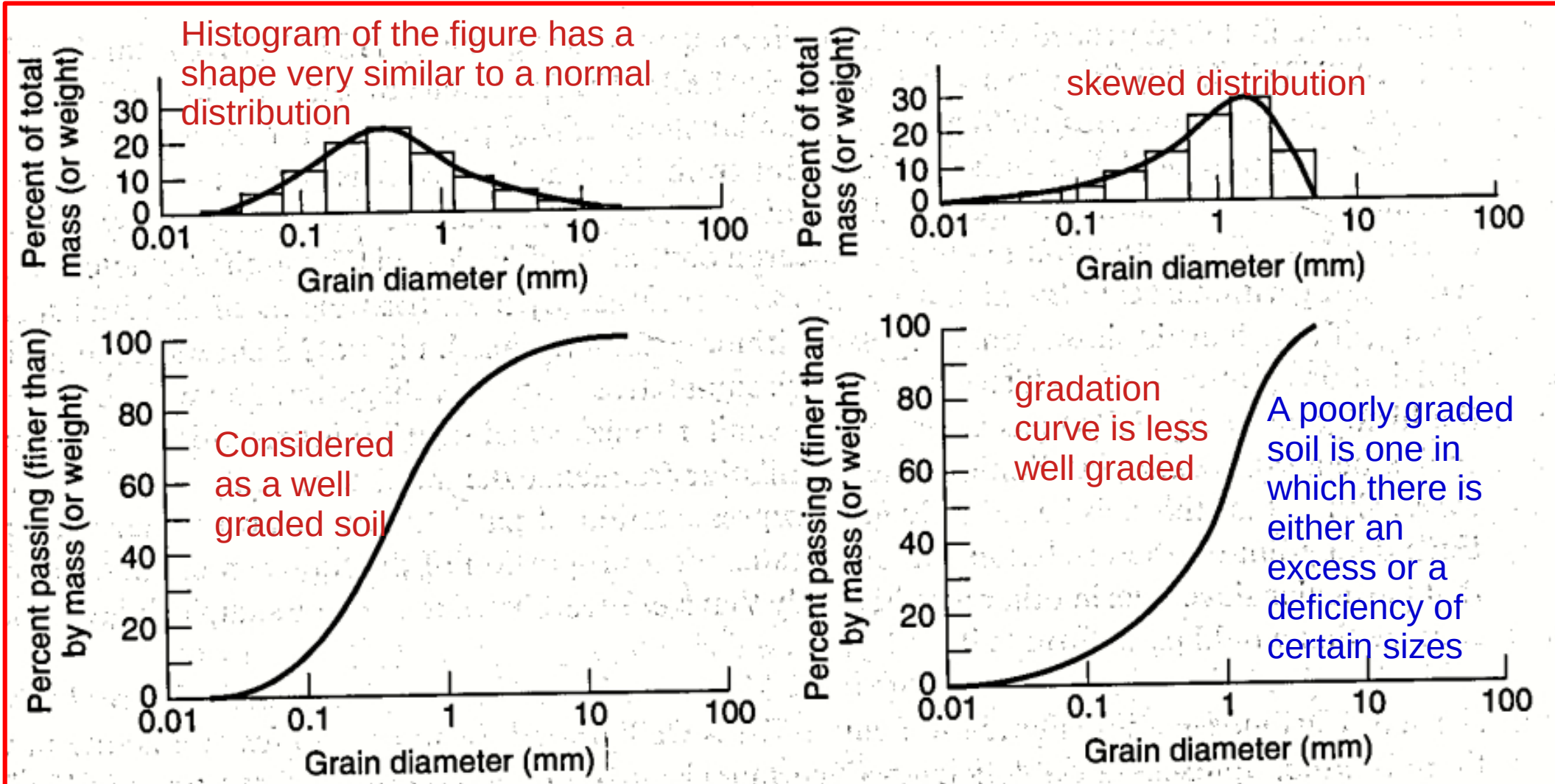
106 μm

200^a

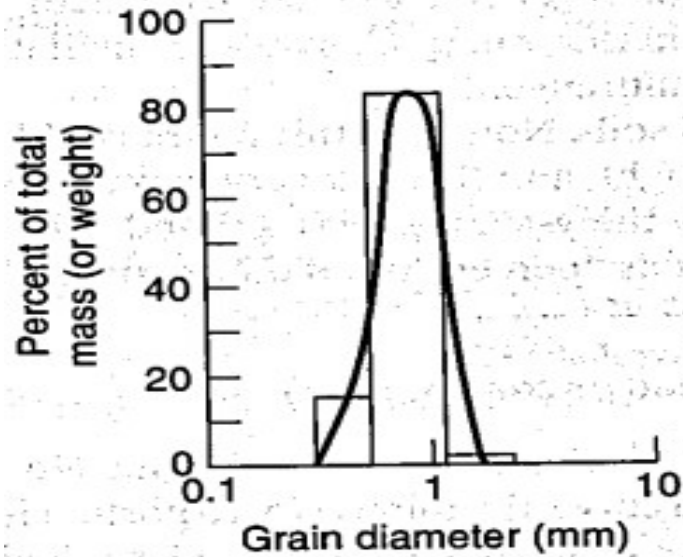
75 μm

Sieves for particle
size analysis

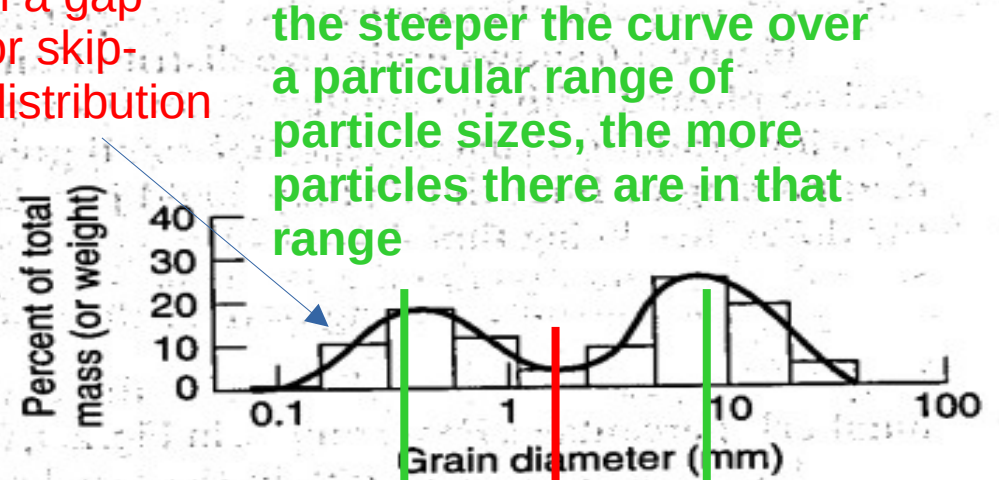
^aUse these sieves to provide a uniform spacing on the grain-size distribution curve.



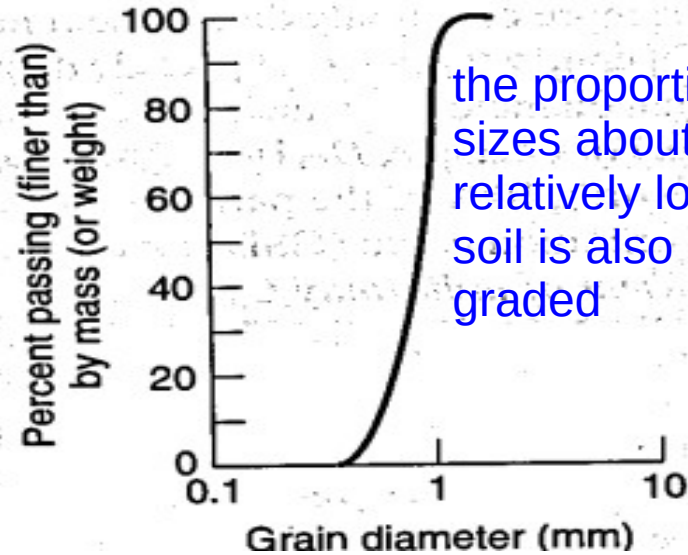
$$\% \text{ passing (or finer than)} = 100 - \% \text{ retained (or coarser than)}$$



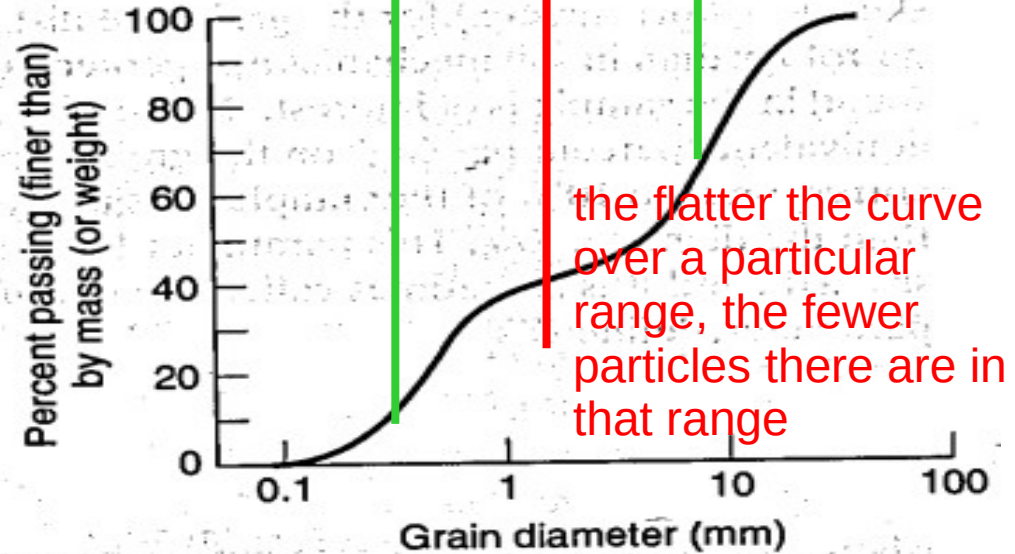
results in a gap
graded or skip-
graded distribution



the steeper the curve over
a particular range of
particle sizes, the more
particles there are in that
range

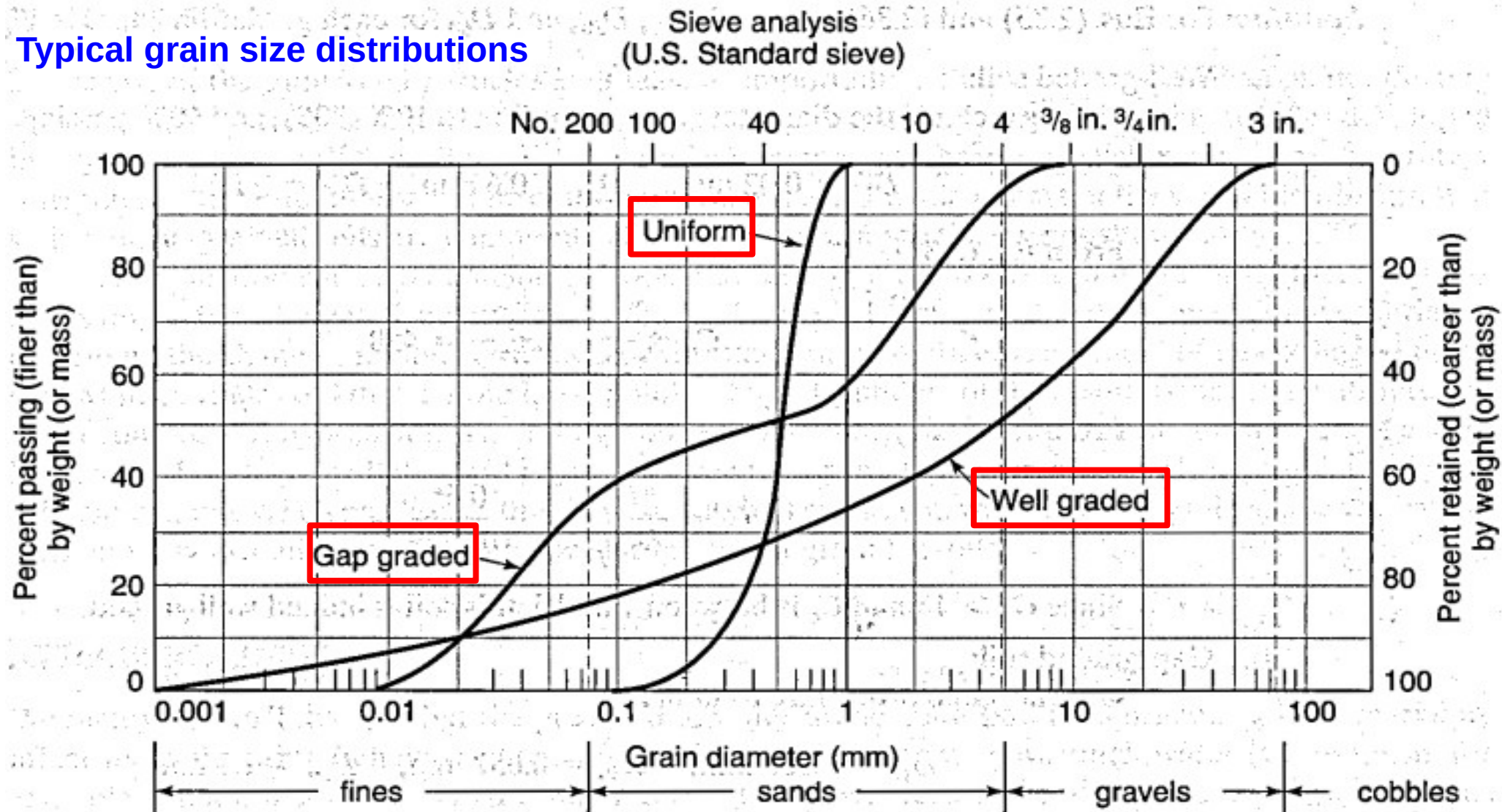


the proportion of grain
sizes about 1-3 mm is
relatively low, and the
soil is also poorly
graded



the flatter the curve
over a particular
range, the fewer
particles there are in
that range

Typical grain size distributions



Use of grain diameters D which correspond to an equivalent "percent passing" on the grain size distribution curve

D_{10} = grain size corresponding to 10% of the sample passing by weight

= 10% of the particles are smaller than the diameter D_{10}

This parameter locates the grain size distribution curve (GSD) along the grain size axis, and it is sometimes called the effective size

$$\text{Coefficient of uniformity} = C_u = \frac{D_{60}}{D_{10}}$$

- $C_u = 1$ would be a soil with only one grain size
- Very poorly graded soils, such as beach sands, have a C_u of 2 or 3
- Very well graded soils may have a C_u of 15 or greater
- As an
- C_u of between 400 and 500; indicates that the sizes range from large boulders down to very fine-grained clay particles.

Coefficient of curvature = $C_c = \frac{D_{30}^2}{D_{10}D_{60}}$

- A soil with a coefficient of curvature between 1 and 3 is considered to be well graded.

As long as :

C_u is > 4 (for gravels)

C_u is > 6 (for sands)

ATTERBERG LIMITS

- the presence of water in the voids of a fine-grained soil can markedly affect its engineering behaviour
- we need to compare or scale the water content against some standard of engineering behaviour

Plasticity is the most conspicuous physical characteristic of clayey soils

- plasticity could be used to distinguish between plastic and non plastic soils i.e. between clays and sand silts (see table in slide-3)
- Finally, plasticity depends on the water content, of a clayey soil

Atterberg limits- the threshold water contents at which certain types of engineering behaviour can be expected.

- These water content boundaries are unique for each soil, but the referenced soil behaviour is the same
- Atterberg limits are the most important items in the description of fine-grained soils.
- They are used in classification such soils,
- and they are useful because they correlate with the engineering properties and engineering behaviour of fine-grained soils.

1. **Upper limit of viscous flow**

2. **Lower limit of viscous flow**

3. **Liquid limit** -lower limit of viscous flow.

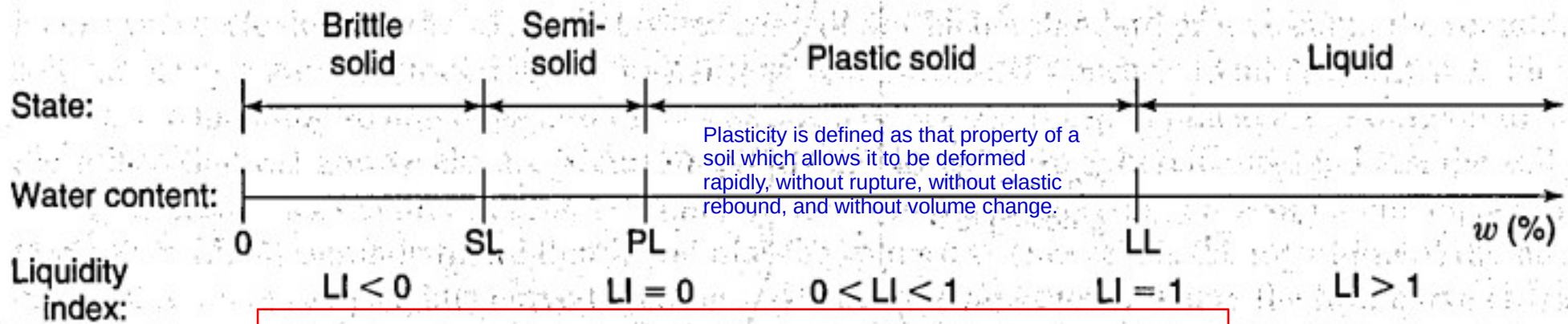
4. **Sticky limit** -clay loses its adhesion to a metal blade.

5. **cohesion limit** -grains ceases to cohere to each other.

6. **Plastic limit**-lower limit of the plastic state

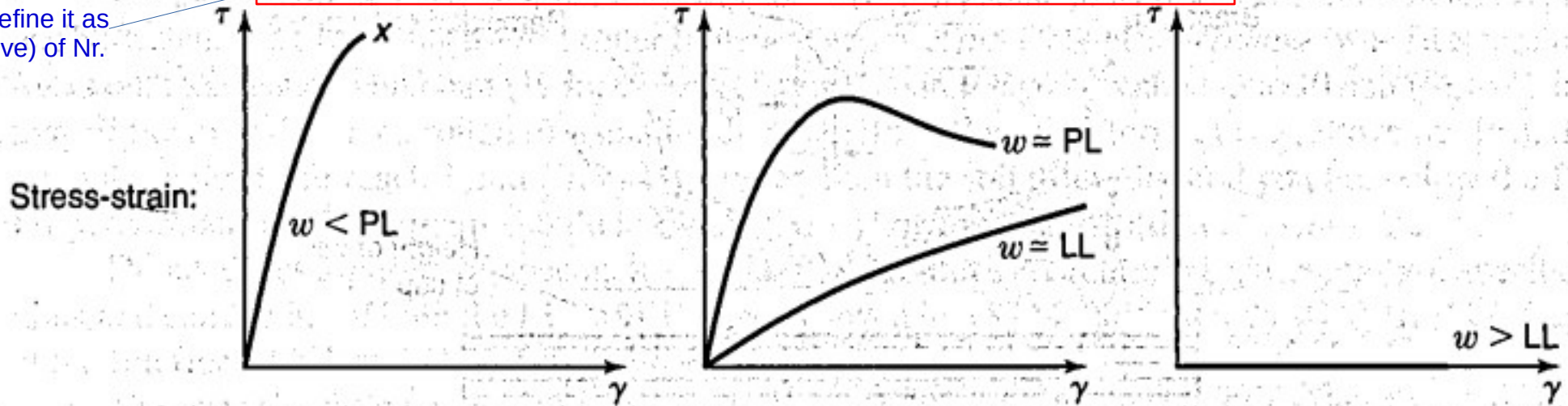
7. **Shrinkage limit**-lower limit of volume change

The sticky and the cohesion limits are more useful in ceramics and agriculture.



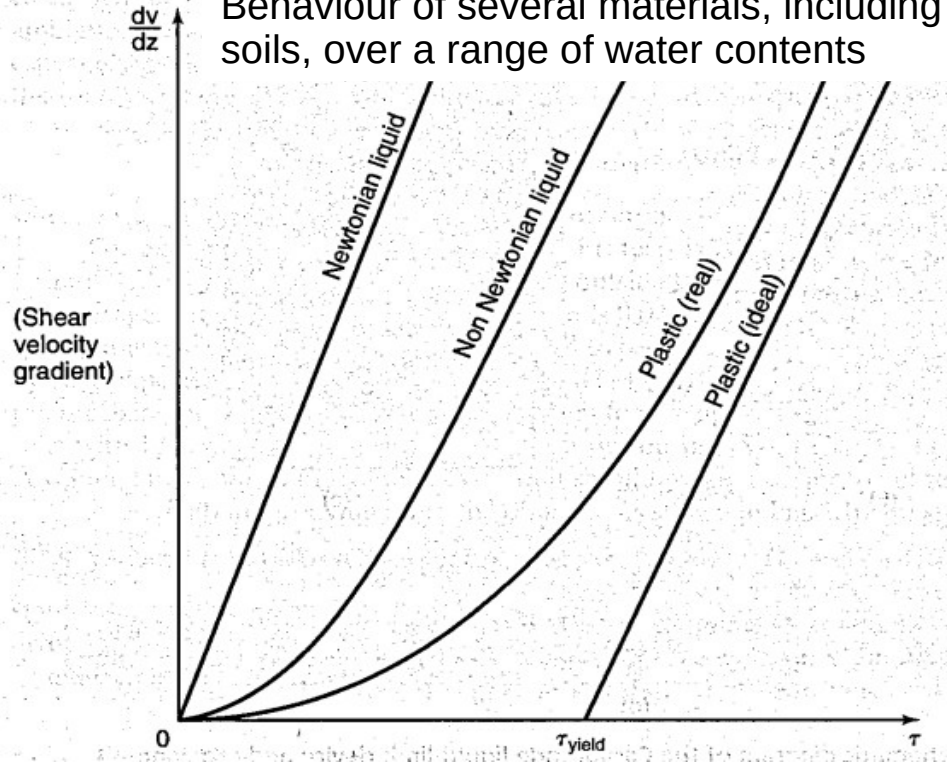
$$LI = \frac{w_n - PL}{PI} \quad w_n = \text{natural water content} \quad PI = LL - PL$$

Few authors define it as (-ve) of Nr.



As the water content increases, the state of the soil changes from a brittle solid to a plastic solid and then to a viscous liquid.

Behaviour of several materials, including soils, over a range of water contents

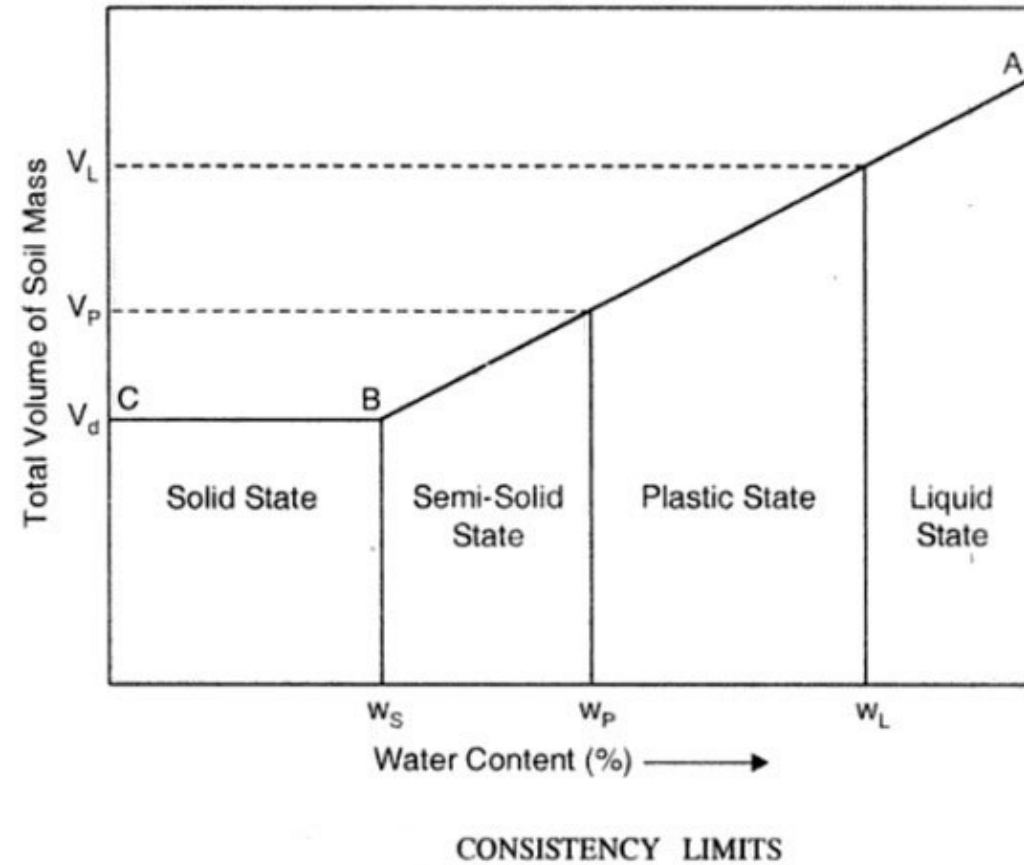


shear velocity gradient is plotted versus the shear stress

- When this relationship is linear, the liquid is called Newtonian; and the slope of the line is, of course, the viscosity
- If the viscosity is not a constant, then the material is; a real or non-Newtonian liquid
- a liquid is defined as a material that cannot support a static shear stress; so, as shown in Fig. when $v = 0$, $T = 0$ for both liquids.

Depending on the water content, it is possible for soils to have a response represented by all these curves except the ideal Newtonian liquid.

Note, too, how different this response is from the stress-strain behaviour of other engineering materials



Plasticity is defined as that property of a soil which allows it to be deformed rapidly, without rupture, without elastic rebound, and without volume change.

Liquid Limit.

Liquid Limit is defined as the minimum water content at percent on dry bases at which the soil changes its state from Liquid to Plastic State.

Plastic Limit.

Plastic Limit is defined as the minimum water content at percent on dry bases at which a soil enters from Semi-Solid State to Plastic State.

Shrinkage Limit.

- Shrinkage limit is defined as, the maximum water content at which a reduction in water content will not cause a decrease in the volume of a soil mass.
- It is lowest water content at which soil can still be completely saturated.

Plasticity Index.

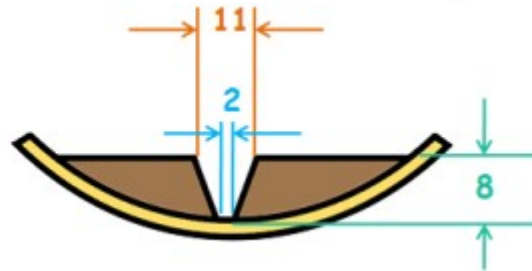
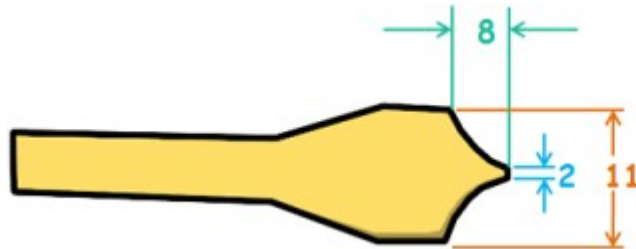
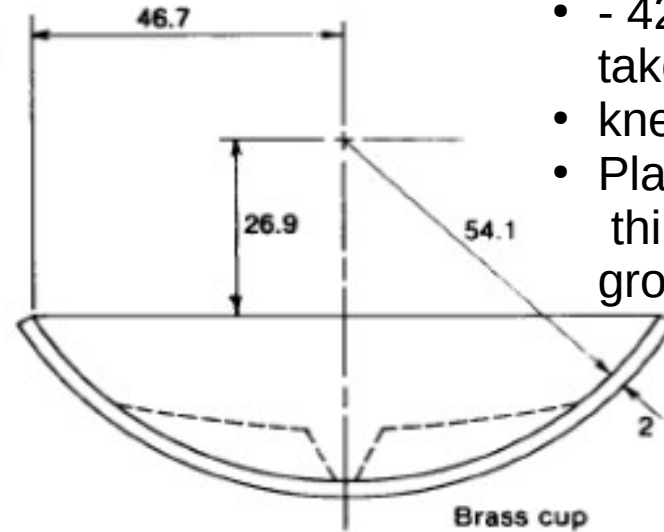
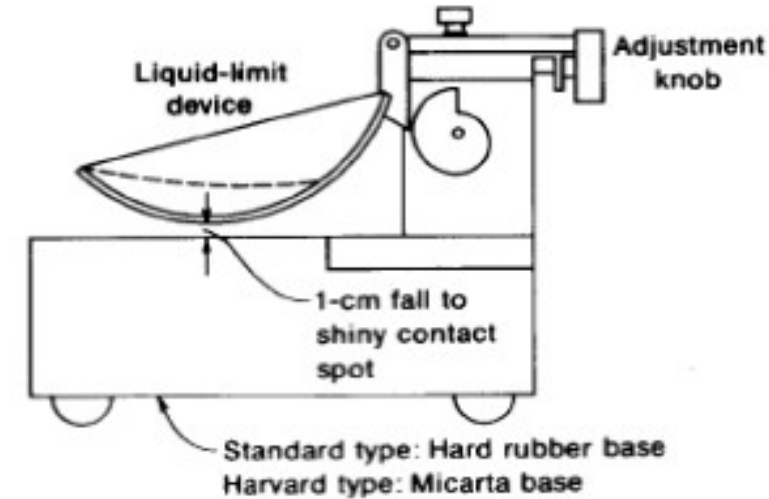
- The range of consistency within which a soil exhibits plastic properties is called plastic range and is indicated by plasticity index.
- The plasticity index is defined as, the numerical difference between the liquid limit and the plastic limit of a soil: $P.I = LL - PL$
- In the case of sandy soils, the plastic limit should be determined first. When plastic limit cannot be determined, the plasticity index is reported as NP (non-plastic).
- When the plastic limit is equal to or greater than the liquid limit, the plasticity index is reported as zero..

Goldschmidt theory of plasticity

- the plasticity is due to the presence of thin scale like particles which carry on their surfaces electro-magnetic charges.
- Water molecules are bi-polar and orient themselves like tiny magnets in the magnetic field next to the surface of the soil particles.
- Water becomes highly viscous near the particles, but as the distance increases, the viscosity of water decreases until at some distance ordinary water exists.
- When enough water is present (corresponding to the plastic state of consistency), the particles are separated by molasses-like water which allows particles to slip past each other to new positions without any tendency to return to their former positions, without change in volume of voids, and without impairing the cohesion.

The correctness of Goldschmidt's theory for the cause of plasticity is evidenced by the fact that clay does not become plastic when mixed with a liquid of non-polarizing molecules like kerosene.

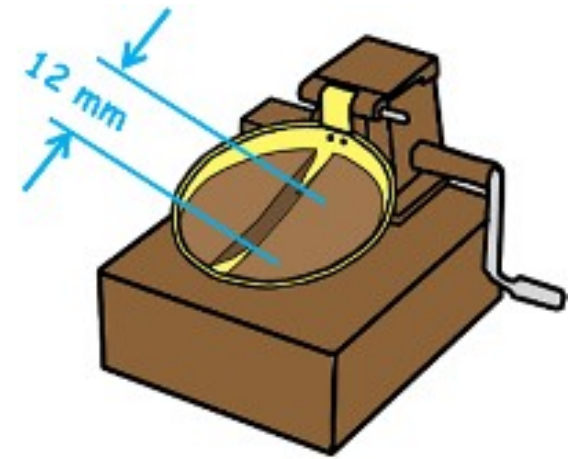
Det. of Liquid Limit Casagrande Method



- - 425 micron IS sieved soil is taken.
- kneaded with water
- Placed in the cup of eqpt. and thin groove is cut using grooving tool.
- apparatus is turned at a rate of 2 revolutions per second, which applies blows to the grooved soil pat.

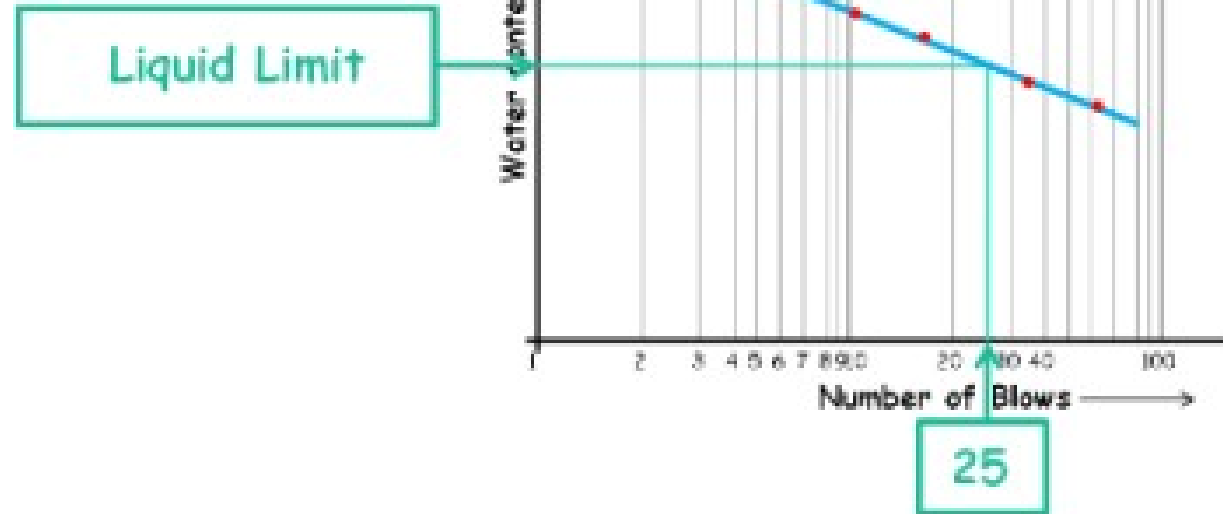
We should take care that the groove should close by flow of soil not by slippage of it. And if slippage occurs then we should discard that soil pat and re-do the whole procedure.

- The number of blows required to close the groove for the length of 12 mm is noted as N1.
- About 10 gm of soil near the closed groove is taken for the water content determination. Water content is determined using any of the water content determination methods. Determined water content is noted as W1.

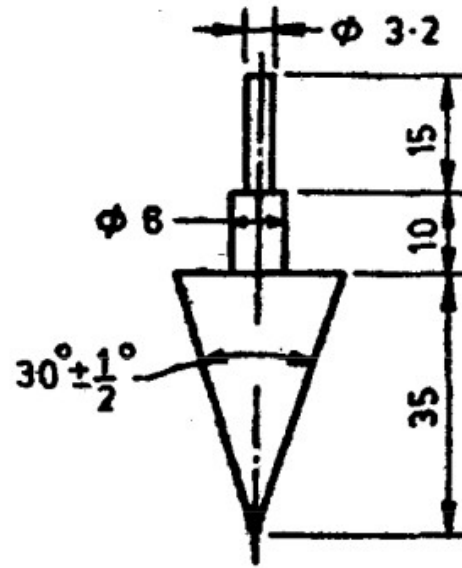


The liquid limit has been defined as the water content of the soil at which the groove closes to 12 mm in 25 number of blows.

- practically, it is difficult to get exactly 25 blows for the 12 mm groove closure, so we conduct the test for different water contents so as to get the number of blows in the range of 10 to 40.



Det. of Liquid Limit Cone Penetrometer Method



All dimensions in millimetres.

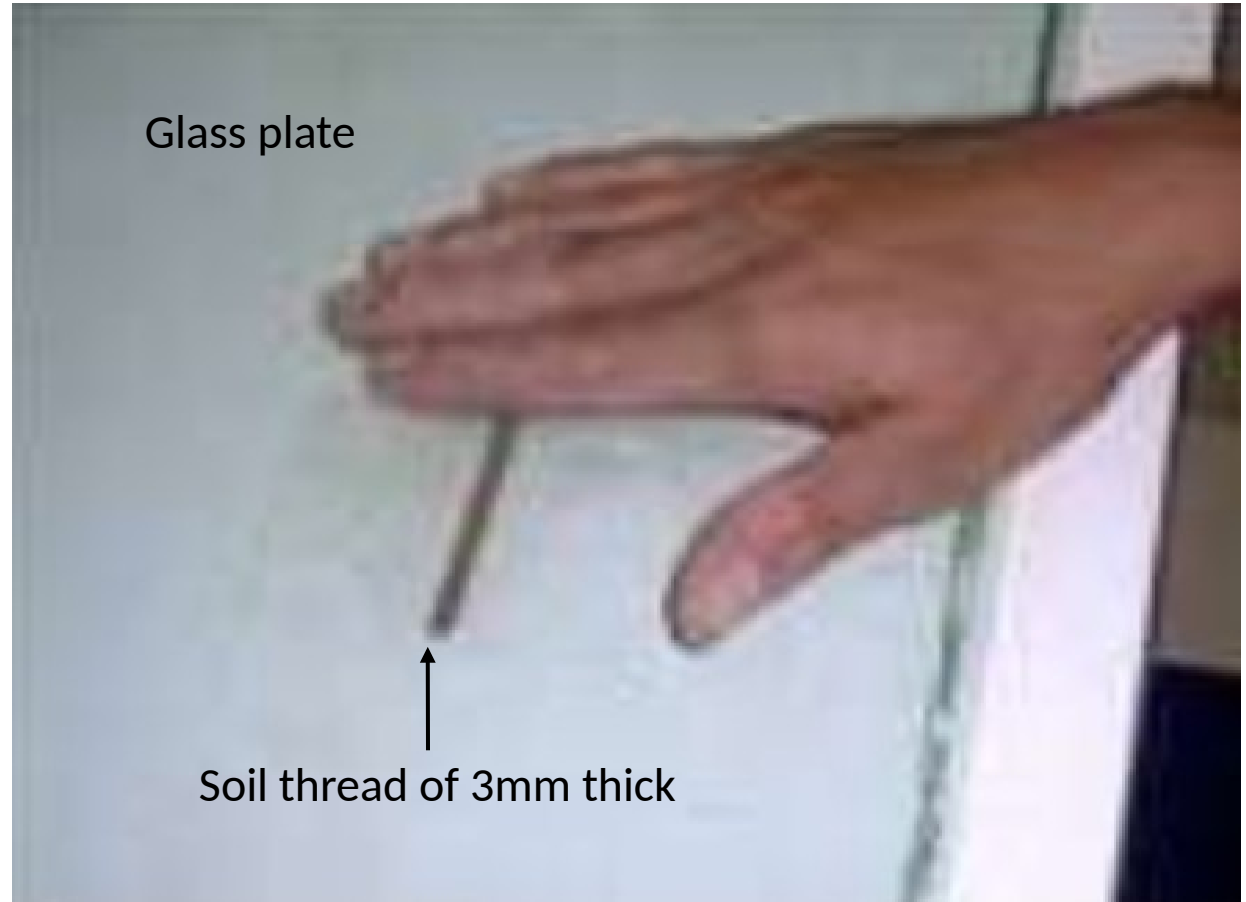
Mass of plunger along with components = 80 g

- 150 g soil of -425 micron size.
- The **penetration of cone after 5 s** noted nearest to mm.
- If the difference lies between 14 and 28 mm, the test is repeated with suitable adjustments to moisture either by addition of more water or exposure of the spread paste on a glass plate for reduction in moisture content.
- Repeated test are conducted for different moisture content and penetrations noted.
- A graph representing **water content on y-axis** and cone penetration on x-axis is prepared. The best fitting line is then drawn.

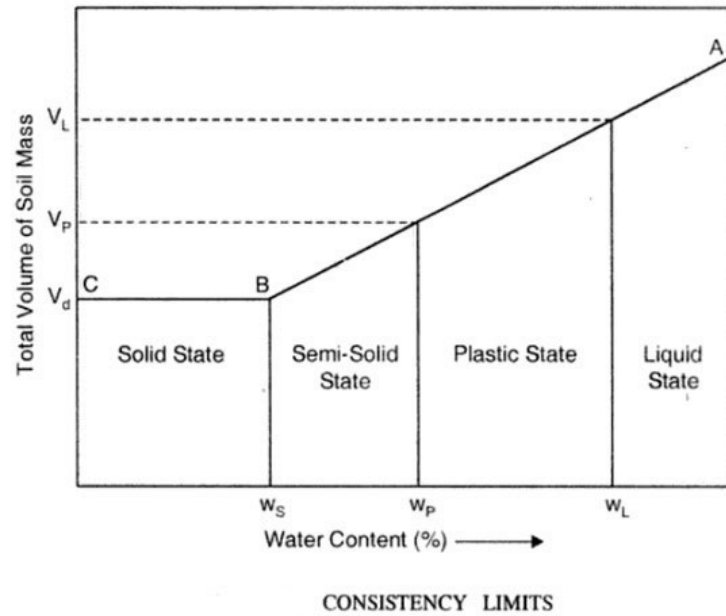
The moisture content corresponding to cone penetration of 20 mm is taken as the liquid limit of the soil.

Determination of Plastic Limit (PL)

- The minimum water content at which a soil will just begin to crumble when it is rolled into a thread of approximately 3 mm in diameter.
- When point is reached where thread is cracking and cannot be re-rolled to 3 mm diameter, collect at least 6 grams and measure water content.
- The test is repeated taking a fresh sample each time. Plastic limit is taken as average of three values.



Determination of Shrinkage Limit of Soil



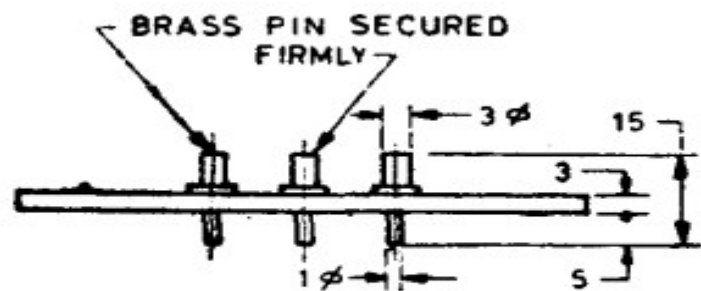
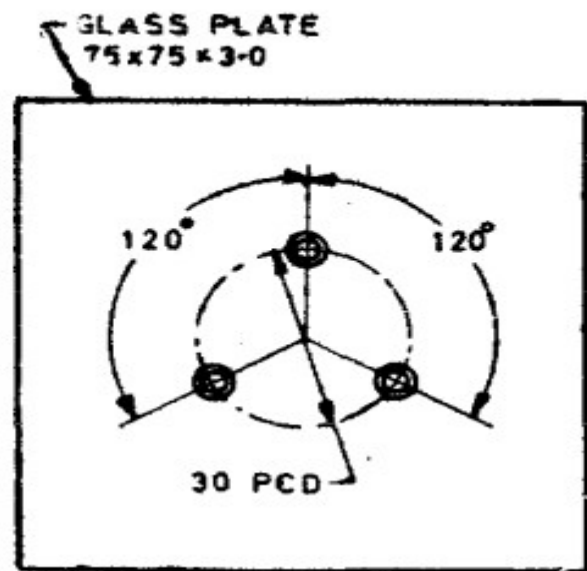
Glass cup – 50 to 55 mm in diameter and 25 mm in height

Glass plates with metal prongs – 75 mm x 75 mm, 3 mm thick.

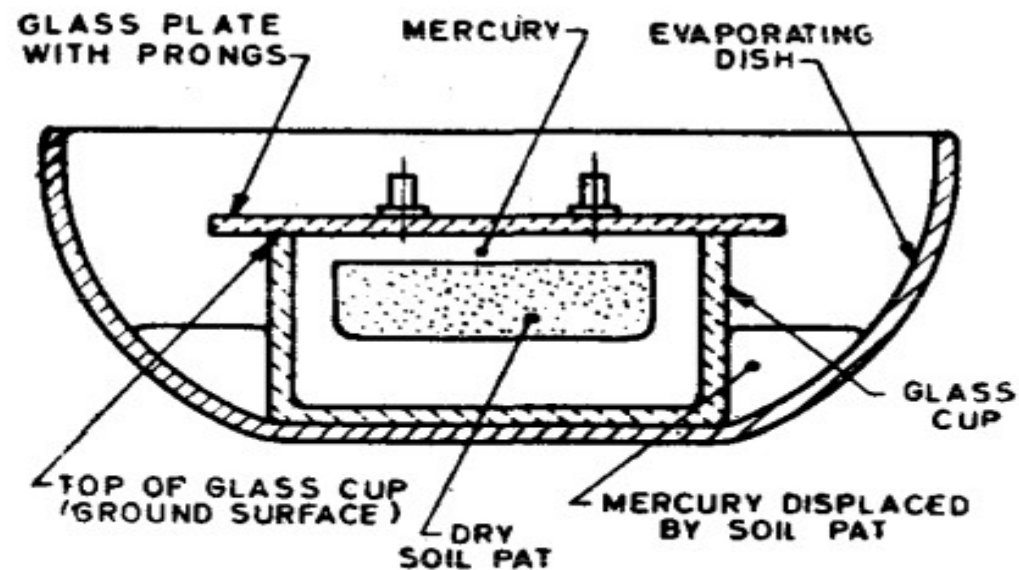
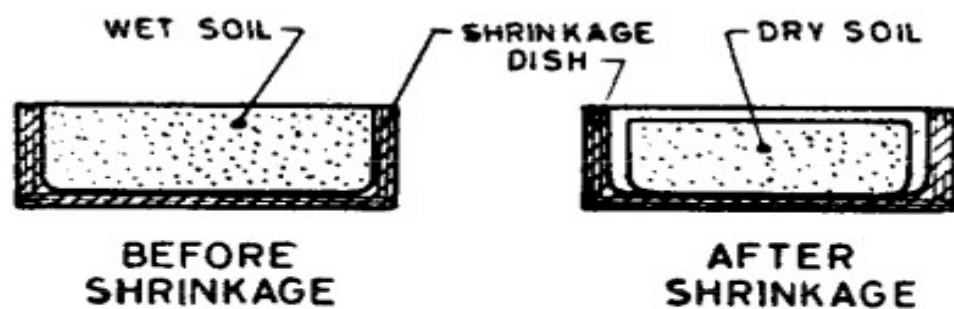
Straight edge -150 mm in length.

Evaporating dish

Shrinkage dish : 45 mm dia. and 15 mm in ht



DETAIL OF GLASS
PLATE WITH PRONGS



METHOD OF OBTAINING
DISPLACED MERCURY

- 1) Clean the shrinkage dish and weigh it (M1). Fill the dish with mercury. Remove the excess mercury by pressing a glass plate over the top of the dish. Weigh the mercury, divide it by the density of mercury, and obtain the volume of the dish which is the volume of the wet soil (V).
- 2) Take about 30 gm of soil, passing the 425 μm IS sieve, in an evaporating dish and thoroughly mix it with water of an amount slightly greater than the liquid limit. The soil-water mixture should be capable of flowing if allowed to drop.
- 3) Coat the inside of the shrinkage dish with a thin layer of grease and fill one-third of the dish with soil-water mixture. Tap the dish on a firm surface, cushioned by several layers of blotting paper, rubber sheet, or similar material. In three operations, completely fill the dish. Strike off the excess soil paste with a straight edge, clean the outside surface, and weigh (M2).
- 4) Dry the soil pat in the air until the color changes from dark to light, and then dry it in a temperature-controlled oven. After drying, cool it in the air and weigh the shrinkage dish and dry the soil pat (M3).

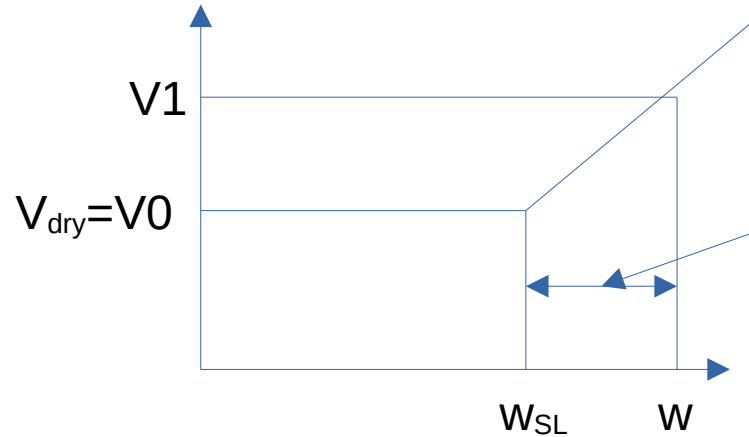
M2-M1 = M = mass of soil

M3-M1 = M0 = oven dried mass of soil (mass of solid)

$$w \% = \frac{M - M0}{M0} \times 100$$

5. Fill the glass cup with mercury and remove the excess mercury by pressing the glass plate with three prongs firmly over the top of the cup.
6. Place the glass cup with mercury in a large evaporating dish and place the dry soil pat on the surface of the mercury.
7. Force the soil pat under the mercury carefully by means of the glass plate with the prongs, so that the soil pat is completely submerged in mercury.
8. Collect the displaced mercury, weigh it, and find its volume, which is the volume of the dry soil pat V_0 .

Shrinkage limit (re-molded soil)

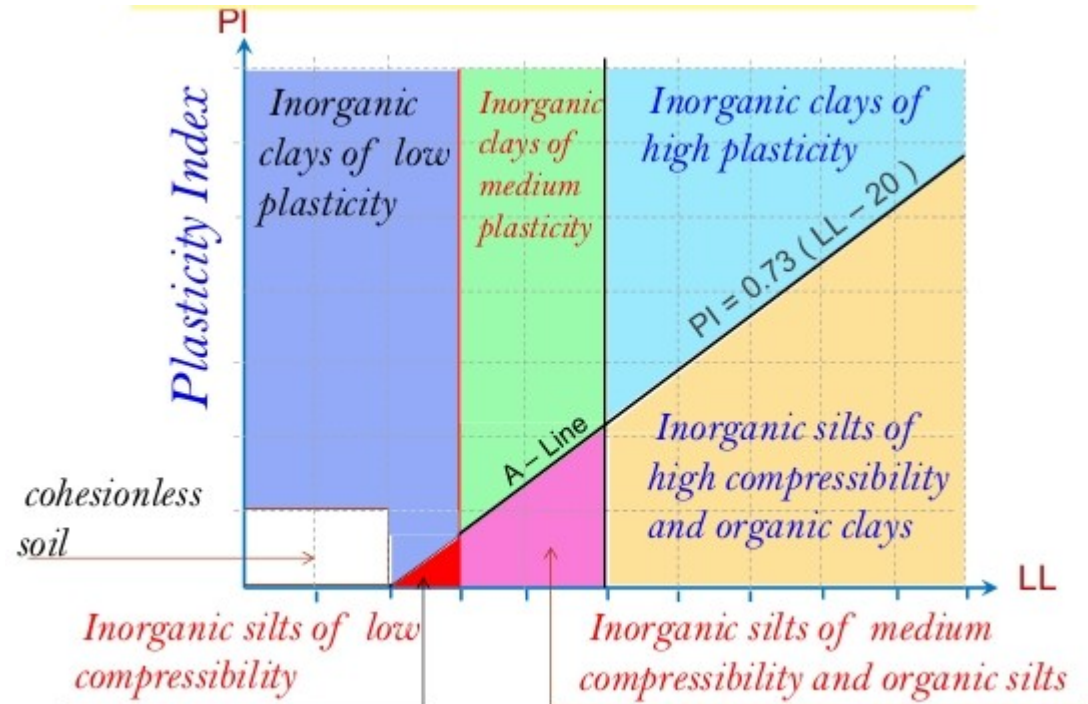
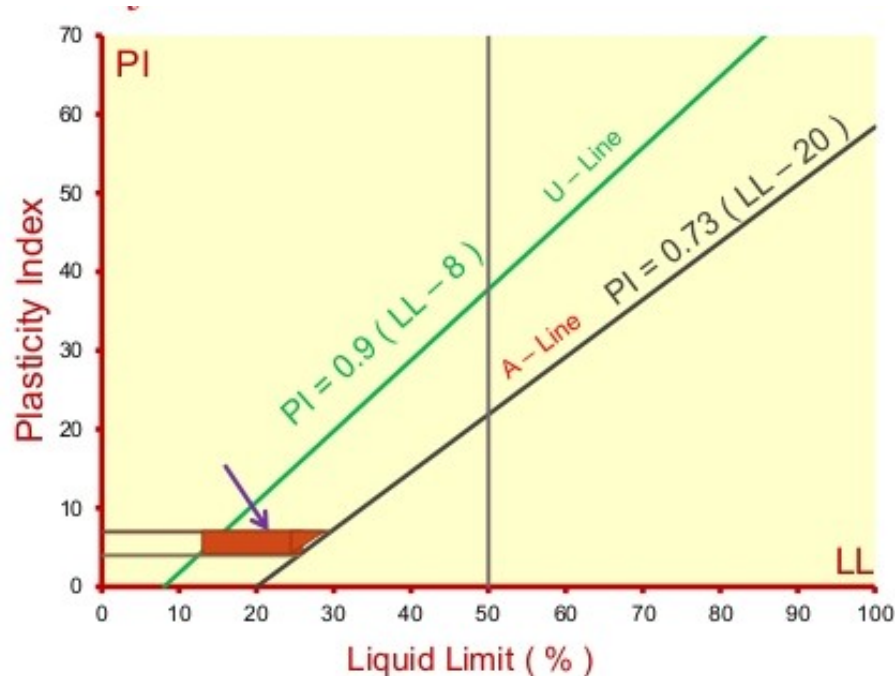


$$SL = w - \frac{(V_1 - V_0)\rho_w}{M_{\text{solid}}} \times 100$$

$$SL = w - \frac{(V_1 - V_0)\rho_w}{M_0} \times 100$$

Plasticity Chart

- Organic clays plot in the same region as inorganic silts of high compressibility (**below the A-line and LL greater than 50**).
- The information provided in the plasticity chart is of great value and is the basis for the classification of fine-grained soils in the Unified Soil Classification System
- The **U-line** is approximately the upper limit of the relationship of the plasticity index to the liquid limit for any currently known soil.



Plasticity Chart

- Three limits are defined based on the water content that causes a change of state.
- The plasticity index defines the range of water content for which the soil behaves like a plastic material .
- The liquidity index gives a measure of strength.
- The soil strength is lowest at the liquid state and highest at the solid state.